

# A Digitalized Clock Drawing Test to Enhance Dementia Screening Intentions in Asian Older Adults

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## Abstract

Dementia poses a global health challenge, with Asia shouldering a significant burden. Despite the benefits of early detection, screening rates among seniors remain low. This paper investigates strategies to enhance dementia screening intentions among Asian older adults through a two-stage study. In Study 1, using a Theory of Planned Behavior-based framework, we identified key barriers, including limited accessibility and financial concerns, which negatively impact dementia screening intentions. These insights informed the development of a digitalized Clock Drawing Test (DigiCDT), a cost-effective and accessible self-screening tool for home and community use. Study 2 evaluated the DigiCDT, demonstrating high acceptability and increased screening intentions when the app was free and accessible. Important insights, such as reduced familiarity with

analog clocks and preferences for community-integrated screening, highlight the need to adapt strategies to cultural differences and technological shifts. This research underscores the potential of digital tools to enhance dementia screening accessibility and participation.

## CCS Concepts

• **Social and professional topics** → **Seniors**; • **Human-centered computing** → *Ubiquitous and mobile devices*.

## Keywords

Dementia Screening, Clock Drawing Test, Screening Intentions, Self-Administered Testing, Digital Health Tools

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## 1 Introduction

Dementia affects approximately 55 million people worldwide, with nearly 10 million new cases reported each year [21]. Asia bears a substantial share of this burden, with China alone accounting for over 16 million cases—nearly 30% of the global total [26]. The disease poses significant personal and economic challenges, leading to cognitive decline and dependency, emotional and physical strain on caregivers, and considerable global economic costs. While there is currently no effective cure for dementia, research suggests that nearly 40% of cases could be prevented or significantly delayed through early screening and intervention [7]. Early detection and intervention plays a crucial role in improving patient outcomes and alleviating the overall burden of the disease by enabling individuals and their families to plan future care, choose appropriate treatments, and implement lifestyle modifications that may help slow cognitive decline [14]. Additionally, early interventions have been shown to ease psychological stress on caregivers [4] and potentially reduce healthcare costs by delaying disease progression and the need for institutionalization [3].

Despite the benefits of early detection, dementia screening rates among seniors remain notably low. A meta-analysis revealed that 61.7% of dementia cases in community settings go undetected, with even higher rates observed in China and India compared to Europe and North America [15]. While macro-level factors such as socioeconomic status, social support, and healthcare system structures contribute to this issue [15], many researchers also emphasize the critical role of individual-level barriers. A key factor identified is the lack of intention among older adults to undergo dementia screening. For instance, a study conducted with 500 seniors in two affluent communities in the U.S. found that approximately half were unwilling to be screened routinely for memory problems [6]. Similarly, a study involving 344 participants aged over 60 in Japan reported that only 32% expressed strong intention to undergo screening, with the majority showing weak interest [24]. These findings highlight the importance of identifying and addressing factors that influence individual intention and behaviour regarding dementia screening to improve screening rates.

In this paper, we aim to develop and evaluate strategies to enhance the intention for dementia screening. We conducted a two-stage investigation focusing on Asian older adults, a demographic less studied in dementia screening research. To better understand the factors influencing dementia screening intentions and behaviours among this demographic, we designed the *Questionnaire for Intention for Dementia Screening* based on the Theory of Planned Behavior [2]. The results from our first study ( $n=31$ ) highlighted perceived behavioral control factors as prominent determinants of intention. Practical barriers, such as limited accessibility to screening services (e.g., lack of knowledge on where or how to access screening, challenges in booking appointments, and distance to healthcare facilities), along with concerns about financial costs, negatively impact the intention to undergo screening. Informed by these insights, we developed a digitalized version of the Clock Drawing Test (CDT), a widely used cognitive test for dementia screening. Designed for deployment on smartphones and tablets, the digitalized CDT (DigiCDT) offers a cost-effective, accessible

solution that supports self-testing and long-term monitoring in home and community environments.

In Study 2 ( $n=13$ ), we evaluated the DigiCDT for usability, acceptability, and its potential to increase screening intentions. The results demonstrated high user acceptance and a notable increase in willingness to undergo screening when the app was free and accessible. Unexpected findings, such as seniors' reduced familiarity with traditional analog clocks, highlighted the need to adapt the DigiCDT design to current digital habits. Additionally, the study emphasized the importance of offering social engagement features and integrating screening into community programs to encourage participation.

## 2 Literature Review

### 2.1 Factors Affecting Intention for Dementia Screening

The intention to undergo dementia screening is influenced by various factors. Using the framework of the Theory of Planned Behavior (TPB) [2], we review the factors reported in prior research based on three key determinants of intention: attitude, subjective norms, and perceived behavioral control.

**2.1.1 Attitude.** Attitude reflects an individual's evaluation of dementia screening, including its risks, benefits, and outcomes. Among the determinants, attitude-related factors are the most widely studied.

One prominent aspect is *awareness and knowledge about dementia*. Greater dementia literacy, including understanding the consequences of dementia and the benefits of early detection, is consistently linked to higher willingness to undergo screening [1, 5, 11, 24]. Awareness of preventive lifestyle factors that can delay cognitive decline also significantly influences attitudes [6, 24]. Another critical theme is the *perception of benefits and harms*. Benefits such as planning for future care, accessing treatment, and reducing uncertainty are key motivators [9–11, 13]. Conversely, perceived harms, including fears of stigma, loss of independence, or distress from a diagnosis, often act as barriers [6, 13]. *Expectations about treatment and outcomes* also influence attitudes. Studies indicate that beliefs regarding the existence and effectiveness of treatments for dementia affect the intention to screen [9, 12].

**2.1.2 Subjective Norms.** Subjective norms refer to the influence of social expectations or pressures, often shaped by significant others such as family, friends, or healthcare providers. *Stigma* is a frequently discussed barrier, as many individuals fear social judgment or being labeled as cognitively impaired [6, 13]. Cultural differences also play a role in shaping subjective norms; for instance, despite a higher reported acceptance of screening, older adults in the UK also reported higher perceived stigma concerns compared to their US counterparts [13]. On the other hand, *social support and familiarity* with dementia facilitate screening. Individuals who know someone with dementia or Alzheimer's disease are more likely to participate, as they better understand the benefits of early detection [10, 22].

**2.1.3 Perceived Behavioral Control.** Perceived behavioral control pertains to an individual's belief in their ability to perform a behavior, which is influenced by both internal and external factors.

Compared to attitudes and subjective norms, perceived behavioral control has been less frequently studied. Only a few studies highlight barriers such as screening accessibility, financial costs and the lack of suitable screening tools [18, 25].

**2.1.4 Summary.** Overall, attitude and subjective norms are the most extensively studied factors influencing dementia screening intentions, with a significant emphasis on awareness, perceived benefits, and stigma. In contrast, perceived behavioral control has received comparatively less attention, despite its critical role in addressing practical barriers to screening. Additionally, most studies focus on populations in North America and Europe [5, 6, 9–13, 22], with relatively few addressing Asian populations [1, 24, 25]. Given the rapidly aging populations in many Asian countries and the difference in culture that may shape attitudes and perceptions, this also represents a considerable gap in the literature. Addressing these gaps is essential for developing comprehensive strategies to enhance dementia screening rates.

## 2.2 Digitalized Dementia Screening Tool

The increasing adoption of digital devices has created new opportunities for dementia screening through digitalized tools, which enable remote screening and automated scoring [28].

Examples include CST [8], DETECT [27], CANS-MCI [19], and NCGG-FAT [17]. These tools are deployed in either web-based environments or portable devices, though they differ in interface design and the cognitive domains they assess. Despite these differences, they share a common focus on enhancing accessibility, streamlining cognitive assessment processes, and minimizing dependence on clinical expertise. Evaluations typically measure reliability, validity, sensitivity, and specificity. CST [8] and DETECT [27] emphasize diagnostic accuracy, while NCGG-FAT [17] focuses on multidimensional assessments with strong reliability. Many tools also prioritize usability and accessibility, ensuring feasibility for diverse populations, including older adults with limited digital literacy [28].

Despite their promise, most evaluations focus on the tools' screening accuracy and usability, assuming participants are already willing to screen. Few studies examine the potential of these tools to improve the intention of individuals to participate in screening, highlighting a gap in existing research.

## 3 STUDY 1: Factors Affecting Intention to Screen

### 3.1 Questionnaire Design

This study aimed to examine the factors influencing the intention to participate in dementia screening among Asian older adults. To achieve this, we conducted a qualitative hybrid study incorporating a questionnaire followed by semi-structured interviews. Our questionnaire design was grounded in the Theory of Planned Behavior (TPB) framework [2], which is commonly employed to explore attitudes, subjective norms, perceived behavioral control, and intention related to a target behavior. While TPB questionnaires are well-established for quantitative studies, we adapted the design to better suit our qualitative methodology and senior participants.

Standard TPB questionnaires often present several challenges in this context. Their comprehensive nature can result in a high

length and time commitment, which may overwhelm participants, particularly older adults. Additionally, the complexity of wording in traditional TPB questionnaires, often characterized by verbose and abstract phrasing, can confuse seniors, especially those with lower levels of educational attainment. Finally, the close-ended nature of standard TPB questionnaires limits the ability to gather in-depth, exploratory insights, which are essential for understanding nuanced perspectives.

To address these limitations, we developed a concise and age-friendly *Questionnaire for Intention for Dementia Screening* comprising four sections (full questionnaire available in **Appendix A**):

- **Demographics (Q1–Q5):** Capturing participant characteristics.
- **Health Screening Knowledge and Behaviors (Q6–Q11):** Assessing baseline knowledge and practices regarding general and dementia screening.
- **Theory Planned Behavior (TPB) Constructs (Q12–Q17):** Investigating attitudes, subjective norms, and perceived behavioral control related to dementia screening. These questions were reformatted into multiple-choice items to reduce cognitive load. For instance: Instead of several Likert-scale questions about subjective norms, participants were asked, “What prevents you from accessing cognitive health screenings? (Select all that apply).” Options included barriers like “Afraid of being mocked during the test” (subjective norms) and “Financial concerns” (perceived behavioral control).
- **Open-Ended Questions (Q18):** Designed to inspire deeper reflection and serve as a springboard for the subsequent semi-structured interviews.

### 3.2 Study Participants and Procedures

We recruited 31 volunteers from senior activity centers and canteens in Singapore (n=19) and mainland China (n=12). Participant demographics included:

- **Age:** 23 participants aged 55–64; 8 participants aged 65 and above.
- **Gender:** 17 males and 14 females.
- **Education:** 4 participants with less than six years of education, 10 with high school education, and 17 with university education or higher.
- **Living Status:** 2 participants lived alone, while 29 lived with family.
- **Living Areas:** 23 participants resided in urban areas, while 8 were from smaller towns or villages.

We conducted five workshops following the TPB framework. Each workshop involved 5–7 participants and was facilitated by two researchers. The procedure included: Step 1. Introduction: Researchers provided an overview of dementia and cognitive decline, the study's purpose, procedures, and instructions for completing the questionnaire. Participants signed the consent forms when they agree to join the study. Step 2. Questionnaire Administration: Participants completed the pre-designed questionnaire. To encourage feeling and idea expressions, they were prompted to “think aloud” while answering. Step 3. Semi-Structured Interviews: Researchers conducted one-on-one interviews based on participants'

questionnaire responses. The interview topics included exploring the reasons behind participants' answers and gathering specific examples or details to support their responses. Participants were also asked to discuss any additional factors or concerns that were not addressed in the questionnaire options. Furthermore, the interviews delved into participants' perceptions of their friends' or family members' opinions about dementia screening. For participants who were caregivers, their beliefs and actions regarding dementia screening were also discussed. Lastly, participants were encouraged to share any further thoughts or personal experiences related to dementia screening, providing a broader understanding of their perspectives.

Open-ended responses and interview transcripts were analyzed using thematic analysis with axial coding, independently conducted by two researchers. Key themes were identified, and participant quotes were selected to illustrate these themes, offering rich, user-centered insights that contextualize the quantitative usability findings and enhance ecological validity. Any discrepancies in coding and theme identification were resolved through discussion to ensure consistency.

### 3.3 Study Results

**3.3.1 Attitudes.** Most participants recalled experiences related to cognitive decline and linked these issues to critical diseases such as dementia. Nearly half had never heard of cognitive health screening before the workshops. However, after learning about it during the workshop, almost all participants recognized its benefits. This demonstrates notable difference from studies in North America and Europe, where participants often expressed mixed perceptions and concerns about potential harms of dementia screening [6, 13]. Despite the high recognition of benefits, 10 out of 31 participants did not consider screening relevant to them, believing they were healthy. A notable bias was overconfidence in their cognitive health, with some perceiving screening as necessary only for high-risk individuals. Others emphasized personal freedom and authority over health decisions. As Participant 28 stated, *"I don't like any public forcing screening... I do not want to take it. If I get dementia, I'll accept it. That is my choice."* These findings suggest the need for a diagnostic approach that is discreet and respects individual decision-making.

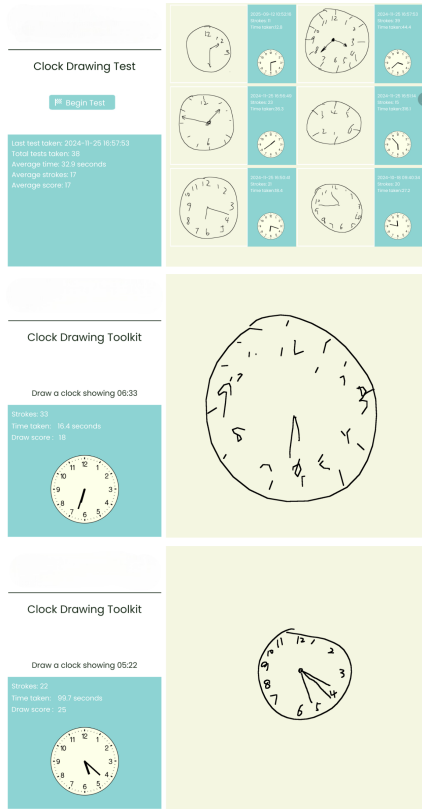
**3.3.2 Subjective Norms.** This section explored participants' perceptions of others' attitudes toward cognitive health screening. Most participants viewed cognitive decline as a common aspect of aging and expressed little shame or concern about potential screening results. Many even joked about forgetfulness with friends and family, as Participant 9 shared: *"My friends and I say, 'I may get dementia,' when we are being silly nearly every day."* This demonstrates a notable difference with studies in North America and Europe, where stigma and fear of being labelled cognitively impaired were significant barriers [6, 13, 16, 20]. Unlike Western populations, where cognitive impairment is often associated with social stigma, Asian participants were less concerned about such labels, instead perceiving cognitive decline as a common aspect of aging. This perspective aligns with collectivist cultural values, where health concerns are typically managed within family networks rather than as individual

responsibilities. Nonetheless, some participants preferred confidential screening methods, as Participant 2 noted: *"If I can do the test at home by myself, I'll do it routinely. Then I can choose whether to share the results or not."* A conformity phenomenon also emerged, with some participants expressing willingness to undergo screening if it were widely promoted. Participant 26 remarked, *"If everybody around me took the screening, I will also take it. But nobody talked about it. I feel weird to do it alone."* Participants' reluctance to seek screening unless widely promoted suggests that social reinforcement plays a critical role in shaping screening behaviors, highlighting the need for community-driven screening initiatives. These findings suggest that while stigma is less of a barrier for Asian older adults, social reinforcement influences screening behaviors, and privacy concerns highlight the need for self-administered options that reduce reliance on in-person visits.

**3.3.3 Perceived Behavioral Control.** This section explored participants' beliefs about their ability to access screenings and whether removing specific barriers might increase their intention to participate. Two key themes emerged from the analysis. The first theme relates to **test accessibility**. Ten out of 31 participants indicated that they lacked information about where or how to access dementia screening. Other factors affecting accessibility include technical barriers, such as difficulties in booking appointments (2 participants), and logistical issues, like the distance to healthcare facilities (2 participants). Seven participants expressed that having screenings available within their local communities would significantly increase their willingness to participate. For example, Participant 19 stated: *"If I don't have to travel too far, and I can do it at the community center."* Another seven participants noted that their intention to screen would improve if the process could be simplified and self-administered at home. For instance, Participant 23 stated: *"...a simple and convenient small program comes with some results and suggestions."* Participants also showed high acceptance of technology-facilitated, self-administered screenings, with two participants specifically mentioning app-based tools as a motivator. The second theme involves **cost considerations**. Six participants identified financial concerns as a barrier, with some suggesting that having screenings covered by healthcare insurance (Participant 17) or provided free of charge (Participant 23) would increase their intention to participate.

## 4 STUDY 2: Development and Evaluation of Digitalized CDT (DigiCDT)

The results of Study 1 highlighted the need for a diagnostic approach that is both discreet and respectful of individual privacy and decision-making. The participants also identified key barriers to dementia screening, including limited accessibility, logistical challenges, and financial costs. To address these needs, we developed a digitalized version of the Clock Drawing Test (CDT), designed for deployment on smartphones and tablets. This self-service tool offers a convenient and self-administered approach, enabling seniors to assess their cognitive health independently in home or community settings. By leveraging technology, the digitalized CDT (DigiCDT) aims to overcome logistical obstacles, enhance accessibility and privacy, and reduce the financial burden associated with traditional



**Figure 1: Screenshots of the Digitalized Clock Drawing Test (DigiCDT). Top: Landing page displaying the history of previously drawn clocks. Middle: Interface for drawing and scoring of a participant. Bottom: Interface for drawing and scoring of another participant.**

screening methods, thereby encouraging greater participation in dementia screening.

The following part is structured as follows: Section 4.1 starts with a brief overview of the traditional paper-and-pencil CDT, followed by an introduction to DigiCDT, including its app design and automatic scoring implementation. Section 4.2 describes the user testing study of DigiCDT, and Section 4.3 presents the key results and analysis.

#### 4.1 Design and Implementation of DigiCDT

The Clock Drawing Test (CDT) is a widely recognized tool for dementia screening due to its simplicity, rapid administration, and cultural neutrality. Subjects are instructed to draw a clock reflecting a specified time, and clinicians assess the drawing based on criteria such as accurate placement of digits and hands. Studies report an average recall and sensitivity of 85% in detecting cognitive deficits [23]. However, the CDT typically requires professional administration and scoring, limiting its suitability for frequent use and long-term monitoring.

Digitalized versions of the CDT enable more proactive, frequent, and cost-efficient dementia screening. To start a session, the test

subject goes to the landing page of DigiCDT and choose to "Begin Test" (Figure 1 left). On the drawing interface, instructions on the left panel indicates the time to be drawn, the test subject draws on the canvas on the right with his/her finger. The drawing of the clock includes the clock face, the numbers, and two clock hands with corresponding positions showing the specified time. After the test subjects completes and clicks the "Done" button, the DigiCDT will perform automatic scoring and display the computing score in the left panel (Figure 1 middle and right).

DigiCDT employs a hybrid approach for segmenting and recognizing clock components (digits, clock face, and hands), combining a spatio-temporal model (DBSCAN + CNN) for analyzing drawing strokes and an image-based deep learning model (Mask R-CNN) for noise-resilient recognition. DBSCAN segments components based on pauses and transitions but switches to Mask R-CNN when excessive noise, such as stray strokes, is detected. Once components are recognized, scoring follows established CDT criteria using a rule-based approach to assess digit placement, clock hand accuracy, and sequence correctness. To address the challenge of limited access to large-scale patient data, synthetic clock drawings were generated for effective model training. Accuracy evaluation with 100 clock drawings from participants aged 21–55 showed a Pearson correlation of 0.763 with expert scores, demonstrating the reliability of DigiCDT.

#### 4.2 User Testing

In Study 1, we invited participants to join further studies, retaining contact details for those interested. Thirteen participants ( $n=13$ ) from the original cohort took part in the user testing study. We conducted one-on-one, face-to-face workshops, where each participant tested the acceptability and usability of the DigiCDT prototype. Each session lasted approximately 30 minutes, and participants received a 15 SGD incentive. After a brief introduction to the prototype and instructions for operation, participants independently tested the DigiCDT while engaging in a "think-aloud" protocol, verbalizing their thoughts and reactions as they interacted with the system. This approach helped capture real-time usability challenges and participants' cognitive processes while completing the test. Their performance, expressions, and the time required to complete the test were recorded.

Semi-structured interviews were conducted post-testing to gather participants' perceptions of the DigiCDT and their intention to use it for dementia screening. The discussions covered their overall impressions, the DigiCDT's potential for cognitive decline screening, ease of use, likelihood of installation, willingness to recommend it to others, and suggestions for improving the design. The guiding questions in the semi-structured interview are available in **Appendix B**. We summarize the key themes from the semi-structured interview as follows.

We analyzed participants' verbalized thoughts, observed behaviors, and post-test interview responses to understand their perceptions of DigiCDT and their intention to use it for screening. The think-aloud data captured real-time interaction challenges, while the semi-structured interviews provided deeper reflections on usability and acceptability. Observed expressions, such as hesitation or frustration, were triangulated with verbal responses to identify

barriers to use. Interview transcripts were analyzed using thematic analysis, following the same approach as in Study 1.

### 4.3 Study Results

**4.3.1 Acceptability and Ease of Use.** All participants successfully completed the DigiCDT independently, demonstrating that its design is intuitive and user-friendly. While some initially struggled with precise touchscreen control, they adapted quickly with minimal practice. This indicates that the DigiCDT has a low barrier to use, even for older adults unfamiliar with digital devices. Overall, participants showed high acceptance of using digital technology for cognitive screening, with 11 out of 13 expressing their willingness to undergo screening on a digital platform.

However, an important and unexpected observation emerged regarding the participants' familiarity with traditional analog clocks. As digital clocks have become more prevalent, many seniors are less accustomed to traditional clock faces with hands. Eight out of 13 participants sought clarification on whether to draw a digital or traditional clock before starting the test. This unfamiliarity was more pronounced among participants with lower educational levels. Two participants, in particular, faced significant challenges in drawing a traditional clock. For example, Participant 20, who had primary-level education, said, *"Everyone uses digital clocks now, and hardly anyone uses those round clocks anymore. I've used them before, but I've forgotten where the hour and minute hands should go."* Similarly, Participant 21, who never attended school, remarked, *"I've never been to school, so I don't know how to tell the hour hand from the minute hand."*

These findings are significant as the Clock Drawing Test (CDT) is a widely adopted dementia screening tool. However, in the digital era, where familiarity with analog clocks is declining, particularly among older adults with lower educational levels, the relevance of this test design warrants re-evaluation. Our results highlight the need to adapt DigiCDT to align with evolving digital habits and cognitive abilities of the target population.

To enhance accessibility, DigiCDT should incorporate onboarding tutorials and adaptive assistance features for users with lower digital literacy. Future iterations could explore alternative interaction methods, such as voice-based instructions or simplified gesture-based input. Furthermore, difficulties in drawing analog clocks, particularly among participants unfamiliar with clock faces, underscore the need for modified test formats or alternative cognitive tasks that better reflect modern technology use while maintaining diagnostic effectiveness.

**4.3.2 Intention for Screening.** For most participants, the primary factor influencing their intention to screen was the financial cost associated with screening tests. When asked, "Would you like to install it?", nearly all participants' first reaction was to inquire about the price and whether they would need to purchase a tablet. Upon learning that the app was free and compatible with their smartphones, they expressed higher intention, with comments such as, *"Why not? It's free and there's no harm"* from Participant 20. Participants who had previously cited transportation or the need for a companion as barriers appreciated the self-service and home-accessibility features of the DigiCDT.

However, some participants noted that financial constraints and economic pressures on older adults could impact their intention to participate in preventive healthcare services. Participant 23 remarked, *"Many older adults face family financial pressures, which are often passed down from their children. Making more money becomes a priority, and they wouldn't care much about their brain health."*

While almost all participants showed increased intention to screen with free options, their continued engagement depended on their perceived accuracy of the test. Participant 19 who demonstrated strong cognitive performance and planning ability, questioned, *"Can it really work? Can you determine my cognitive health just by drawing something so simple? It doesn't feel reliable."*

Another significant observation was the preference for integrating cognitive screening into existing community programs. For instance, Participant 19 suggested, *"Organize everyone together in community activities, like group exercises, and include this screening as part of it—it would make people more willing to do it."* Similarly, Participant 29 proposed, *"Include these game-like tests in nursing homes or community activities."* One participant expressed a clear preference for self-administered tests but believed that in-person screenings might be better for others. Participant 30 commented, *"I don't like being pitched to, especially through expert seminars. I want to screen myself, but I think most seniors don't know about these tests, so door-to-door screening should be done."* Others were indifferent between self-administered or community-based screening, indicating flexibility in their preferences.

These observations suggest that while seniors demonstrate high acceptability of digital screening devices, many still prefer some form of social engagement. This insight provides clear guidance for promoting digital screening tools, such as offering both self-administered and community-based versions to accommodate different preferences and needs.

## 5 Conclusions and Limitations

Our first study examined factors influencing dementia screening intentions among Asian older adults, identifying key barriers such as limited accessibility, logistical challenges, financial concerns, and preferences for privacy in screening decisions. Unlike studies in North America and Europe, where stigma and fear of cognitive impairment are identified as a significant barrier, Asian participants tended to view cognitive decline as a normal part of aging and reported fewer concerns about stigma. This highlights important cultural differences that influence dementia screening behaviors, as perceptions of aging and cognitive health vary across regions.

Based on these findings, we developed a digitalized Clock Drawing Test (DigiCDT) to address these barriers by offering a discreet, accessible, and cost-effective solution for dementia screening. Study 2 demonstrated that the DigiCDT was well-accepted by older adults, with most participants expressing a willingness to undergo screening when the app was free and easy to access. The study also revealed important insights, such as seniors' reduced familiarity with analog clocks due to the prevalence of digital clocks, and a preference for integrating screening into social or community programs to encourage participation. These results suggest that future dementia screening strategies should account for cultural and technological shifts, such as declining familiarity with analog

tools, and prioritize flexibility in screening formats—offering both self-administered and community-based options.

While the findings provide valuable insights, the small sample size may limit generalizability. Future iterations of the DigiCDT should focus on integrating feedback, educational resources, and social features to enhance its functionality and user experience. Despite these limitations, this work highlights the potential of digital tools to overcome barriers to dementia screening and improve accessibility for older adults.

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## A Study 1 - Questionnaire for Intention for Dementia Screening

This *Questionnaire for Intention for Dementia Screening* comprises four section and eighteen questions.

- Demographics (Q1–Q5): Capturing participant characteristics
- Health Screening Knowledge and Behaviors (Q6–Q11): Assessing baseline knowledge and practices regarding general and dementia screening.
- TPB Constructs (Q12–Q17): Investigating attitudes, subjective norms, and perceived behavioral control related to dementia screening.
- Open-Ended Questions (Q18): Designed to inspire deeper reflection and serve as a springboard for the subsequent semi-structured interviews.

In this appendix, we provide the full questionnaire in Table 1.

**Table 1: Questionnaire for Intention for Dementia Screening**

|                                                                                                                                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Questionnaire                                                                                                                                                                                                                                                 |
| <b>Section 1: Demographics</b>                                                                                                                                                                                                                                |
| Q1. What is your age group?                                                                                                                                                                                                                                   |
| <input type="radio"/> 55-64<br><input type="radio"/> 65-74<br><input type="radio"/> 75 and above                                                                                                                                                              |
| Q2. What is your gender?                                                                                                                                                                                                                                      |
| <input type="radio"/> Male<br><input type="radio"/> Female                                                                                                                                                                                                    |
| Q3. What is your highest level of education completed?                                                                                                                                                                                                        |
| <input type="radio"/> Not completed primary school<br><input type="radio"/> Primary school<br><input type="radio"/> Secondary school<br><input type="radio"/> High school<br><input type="radio"/> College / university<br><input type="radio"/> Postgraduate |
| Q4. Where do you currently reside?                                                                                                                                                                                                                            |
| <input type="radio"/> Urban area<br><input type="radio"/> Small towns<br><input type="radio"/> Rural area                                                                                                                                                     |
| Q5. Do you live alone or with others?                                                                                                                                                                                                                         |
| <input type="radio"/> Alone<br><input type="radio"/> With family members<br><input type="radio"/> In a care facility<br><input type="radio"/> Others: _____                                                                                                   |
| <b>Section 2: Health Screening Knowledge and Behaviors</b>                                                                                                                                                                                                    |
| Q6. Do you take physical screening?                                                                                                                                                                                                                           |
| <input type="radio"/> Never<br><input type="radio"/> Sometimes<br><input type="radio"/> Routine checking                                                                                                                                                      |
| Continue on next page                                                                                                                                                                                                                                         |

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Questionnaire (continued)

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**Section 2: Health Screening Knowledge and Behaviors (continued)**

Q7. What items do you check during your physical screening? (select all that apply)

- Basic items such as body fat, blood glucose and so on
- Cardiovascular and cerebrovascular
- Blood tests
- Tumor markers
- Others: \_\_\_\_\_

Q8. Have you ever experienced memory issues such as forgetfulness, difficulty finding your belongings, or trouble focusing? If yes, can you share an example of such an experience?

Q9. Are you aware of any potential consequences or outcomes associated with a decline in cognitive ability? If so, can you share what you know?

Q10. Have you ever heard about cognitive health screenings?

- Yes
- No

Q11. Have you ever undergone a cognitive health screening?

- Yes → **complete Q12 - Q15**
- No → **complete Q12 - Q13 and Q16 - Q17**

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**Section 3: Theory of Planned Behavior (TPB) Constructs**

Q12. In your opinion, who do you think should undergo cognitive health screenings? (select all that apply)

- Somebody who has cognitive declination symptoms
- Patients with chronic diseases such as the Parkinson's Disease
- People with a family history
- The older adults
- Everybody
- Others: \_\_\_\_\_

Q13. Do you feel that cognitive health screenings are necessary?

- Strongly agree
- Agree
- Neutral
- Disagree
- Strongly disagree

Q14. Do you recall how you felt before and during your cognitive health screening? (Select all that apply)

- Fear
- Embarrassment
- Anxiety
- Curiosity
- Indifferent (I don't care)
- Others: \_\_\_\_\_

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Continue on next page

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Questionnaire (continued)

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**Section 3: Theory of Planned Behavior (TPB) Constructs (continued)**

Q15. Do you feel that undergoing cognitive testing has had a negative impact on your self-esteem?

- Strongly agree
- Agree
- Neutral
- Disagree
- Strongly disagree

Q16. What prevents you from participating in cognitive health screenings? (Select all that apply)

- Never heard about this testing
- No idea about how or where to take the testing
- I do not need to do it
- Financial concerns
- Difficulty in booking appointments
- Distance to healthcare facilities
- Lack of transportation support
- Physical mobility issues
- Lack of companion or caregivers
- Afraid of bad results
- Afraid about being mocked during the test

Q17. What would increase your intention to participate in a cognitive health screening? (Select all that apply)

- Expert advice about the necessity via TV, newspaper, or internet
- Positive stories from friends / family
- Free testing
- Convenient locations e.g. community centre or home visit
- Self-administered screenings
- Analysis based on my daily routine data
- Others: \_\_\_\_\_

Q18. If you were to design a widely accessible method for all elderly individuals to undergo cognitive health screenings, what approach or solution would you suggest?

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## B Study 2 - User Testing

Semi-structured interviews were conducted post-testing to explore participants' perceptions of the DigiCDT and their intention to use it for dementia screening. The discussions focused on their overall impressions of the DigiCDT, its potential for cognitive decline screening, ease of use, likelihood of installation, willingness to recommend it, and suggestions for improving its design.

Interviewers began with the following guiding questions, tailoring follow-up questions based on participants' responses to encourage deeper insights:

- Do you like the game? Why or why not?
- Do you believe it can help in cognitive decline screening?
- Would you install it if it were free in the app store?
- Do you believe everyone can learn to use it easily?
- Would you recommend it to others?
- Do you have further comments about the game design?